

Idealized numerical simulations of a 2D flow over a bell-shaped hill

Tutor: Simona Rinaldi
simona.rinaldi7@unibo.it

Laboratory of Complex Terrain
Meteorology course
Prof. L.S. Leo

Bologna, 11/12/2025

Objectives

The aim of this laboratory experience is to study the behavior of flows over complex terrain.

We focus on the case of a 2D flow with constant wind over a small ridge, in different configurations of the atmospheric stability, and varying wind velocity and hill height.

We will do that performing idealized numerical simulations using the Weather Research and Forecasting (WRF) model.

Objectives

By the end of this laboratory, you will be able to:

- Familiarize with the main physical mechanisms controlling flow over orography;
- Gain practical experience in configuring, compiling, and running idealized WRF cases;
- Visualize and interpret model outputs.

Introduction

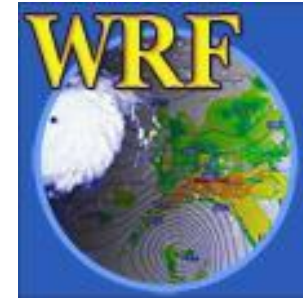
- Flow over a hill • WRF Model

Weather Research and Forecasting (WRF) model

A state-of-the-art mesoscale numerical weather prediction system designed for both atmospheric research and operational forecasting applications.

The model serves a wide range of meteorological applications across scales from tens of meters to thousands of kilometers. For researchers, WRF can produce simulations based on actual atmospheric conditions (i.e., from observations and analyses) or idealized conditions.

Current version is 4.6



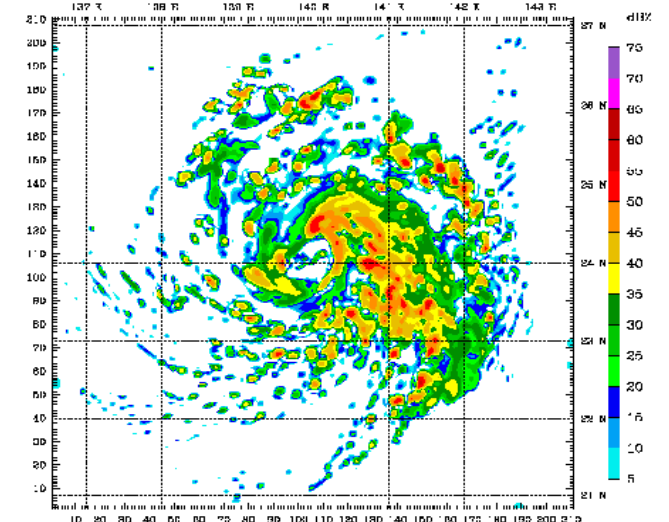
Introduction

- Flow over a hill • WRF Model

Weather Research and Forecasting (WRF) model

What can WRF be used for?

- Atmospheric physics/parameterization research
- Case study research
- Real-time NWP and forecast system research
- Regional climate and seasonal time-scale research
- Coupled chemistry applications
- Global simulations
- Idealized simulations at various scales (convection, baroclinic waves, LES)

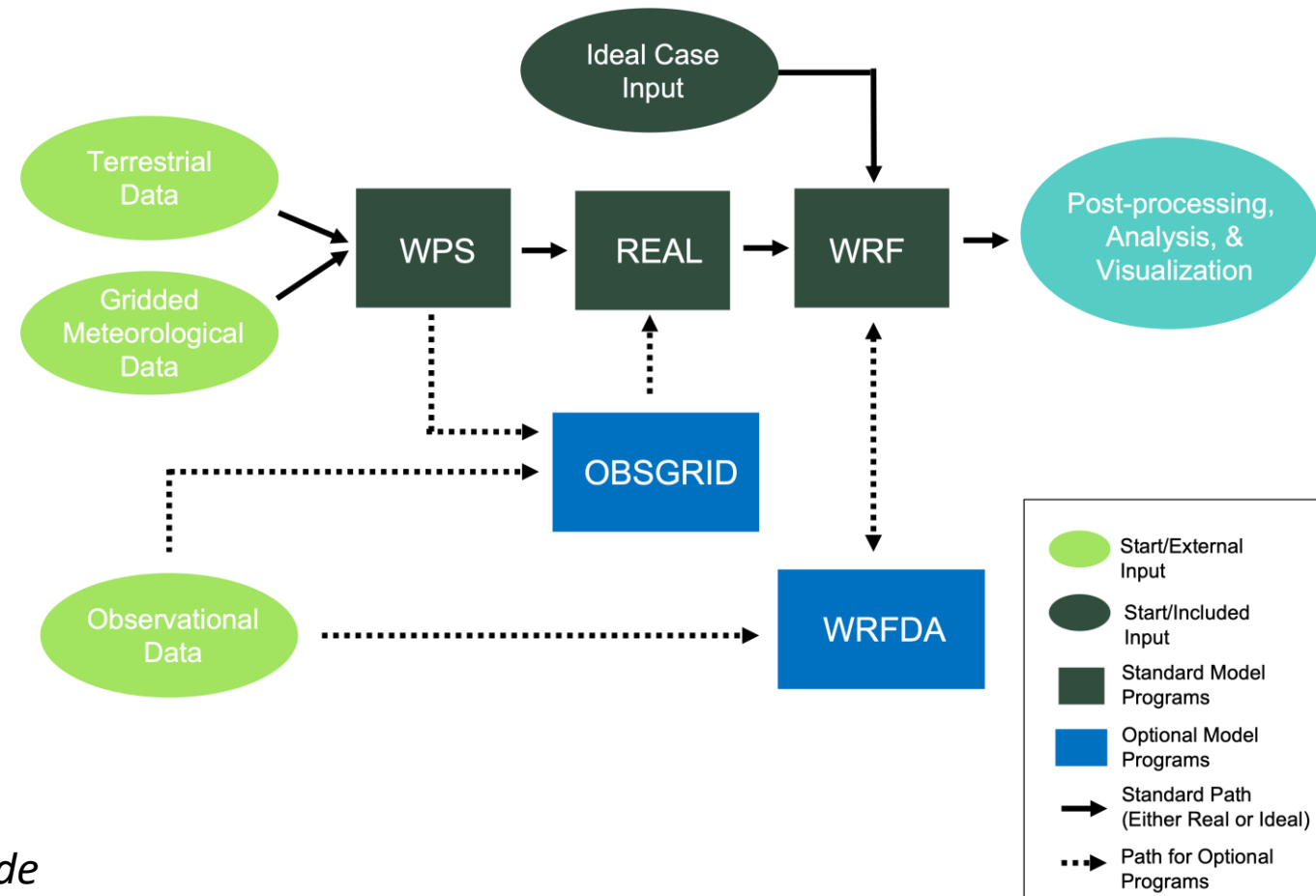


[@Atmoz](#)

Introduction

- Flow over a hill • WRF Model

WRF Modeling System Flow Chart

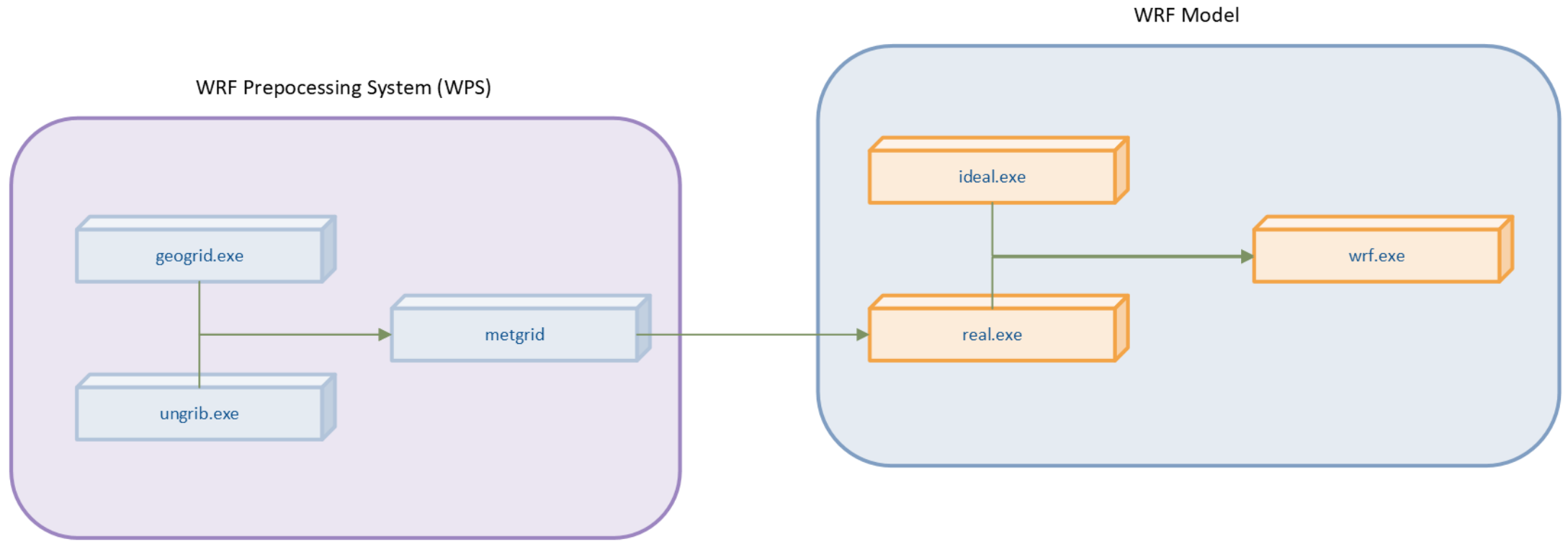


From WRF User Guide

Introduction

- Flow over a hill • WRF Model

WPS and WRF Program Flow



Introduction

- Flow over a hill • WRF Model

- The equation set for ARW is **fully compressible, Eulerian and nonhydrostatic** with a run-time hydrostatic option.
- It is conservative for scalar variables.
- The model uses terrain-following, **hybrid sigma-pressure vertical coordinate** with the top of the model being a constant pressure surface.
- The horizontal grid is the **Arakawa-C grid**.
- The time integration scheme in the model uses the third-order Runge-Kutta scheme, and the spatial discretization employs 2nd to 6th order schemes.
- The model supports both idealized and real-data applications with various lateral boundary condition options.
- The model also supports one-way, two-way and moving nest options.

Introduction

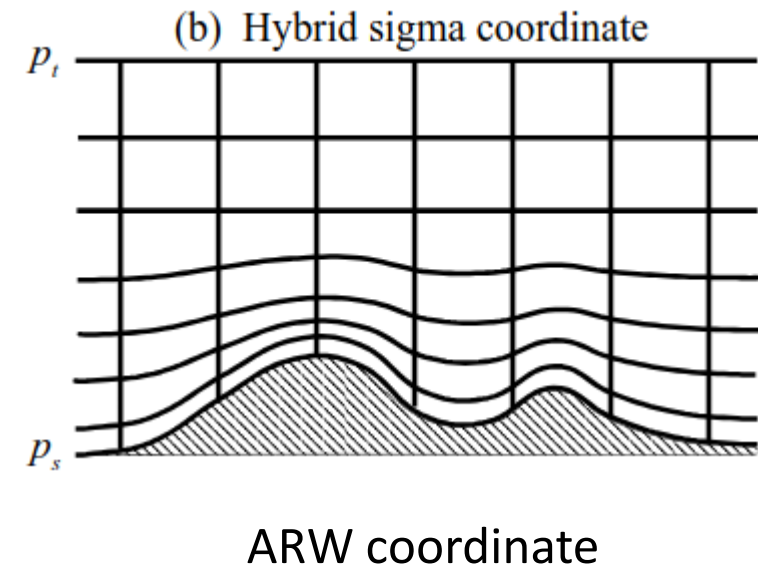
- Flow over a hill
- WRF Model: Vertical coordinates

The ARW v4 uses **hybrid sigma-pressure vertical coordinate** defined as

$$p_d = B(\eta)(p_s - p_t) + [\eta - B(\eta)](p_0 - p_t) + p_t,$$

p_0 : reference sea-level pressure.

$B(\eta)$: defines the relative weighting between the terrain-following sigma coordinate and a pure pressure coordinate, such that η corresponds to the sigma coordinate for $B(\eta) = \eta$ and reverts to a hydrostatic pressure coordinate for $B(\eta) = 0$. To smoothly transition from a sigma coordinate near the surface to a pressure coordinate at upper levels, $B(\eta)$ is defined by a third order polynomial.



Introduction

- Flow over a hill
- WRF Model: Vertical coordinates

The ARW v4 uses **hybrid sigma-pressure vertical coordinate**.

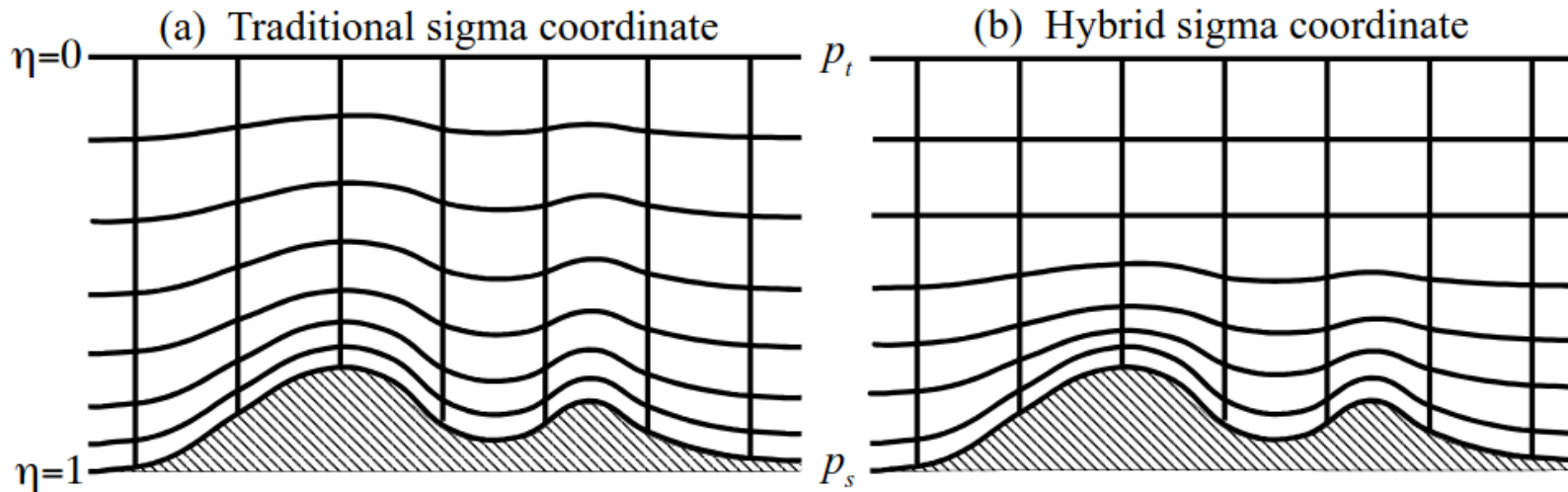


Fig. 2.1 of WRFv4 Technical notes (see references)

Introduction

- Flow over a hill
- WRF Model: Equations

WRF ARW solves the **fully compressible, non-hydrostatic equations of motion** with a terrain-following hydrostatic pressure coordinate system, written in flux form. For simplicity, using Cartesian coordinates and neglecting the Coriolis effect, the WRF model solve the followings:

Equation of state

$$p = \rho R_d T;$$

Mass conservation

$$\frac{\partial \rho}{\partial t} + \frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} + \frac{\partial W}{\partial z} = 0;$$

$$\frac{\partial U}{\partial t} + c_p \Theta \frac{\partial \pi}{\partial x} = -\frac{\partial Uu}{\partial x} - \frac{\partial Vu}{\partial y} - \frac{\partial Wu}{\partial z} + F_x,$$

Conservation of momentum

$$\frac{\partial V}{\partial t} + c_p \Theta \frac{\partial \pi}{\partial y} = -\frac{\partial Uv}{\partial x} - \frac{\partial Vv}{\partial y} - \frac{\partial Wv}{\partial z} + F_y,$$

$$\frac{\partial W}{\partial t} + c_p \Theta \frac{\partial \pi}{\partial z} + g\rho = -\frac{\partial Uw}{\partial x} - \frac{\partial Vw}{\partial y} - \frac{\partial Ww}{\partial z} + F_z;$$

Conservation of energy

$$\frac{\partial \Theta}{\partial t} + \frac{\partial U\theta}{\partial x} + \frac{\partial V\theta}{\partial y} + \frac{\partial W\theta}{\partial z} = \rho Q.$$

$$U = \rho u, \quad V = \rho v, \quad W = \rho w, \quad \Theta = \rho \theta,$$

(u, v, w) are the velocity components

θ is the potential temperature

T is the absolute temperature

π is the Exner function

F_x, F_y, F_z are the friction terms

A more detail description in WRFv4 Technical notes

Introduction

- Flow over a hill
- WRF Model: Spatial discretization

The spatial discretization in the ARW solver uses a **C grid staggering** for the variables.

Normal velocities are staggered one-half grid length from the thermo-dynamic variables.

The variable indices, (i, j, k) indicate variable locations with $(x, y, \eta) = (i\Delta x, j\Delta y, k\Delta \eta)$.

u , v , and w are defined as u points, v points, and w points, respectively, while points where θ is located as mass point.

Moisture variables q_i and the coordinate metric μ are defined at the mass points on the discrete grid, while the geopotential ϕ that is defined at the w points.

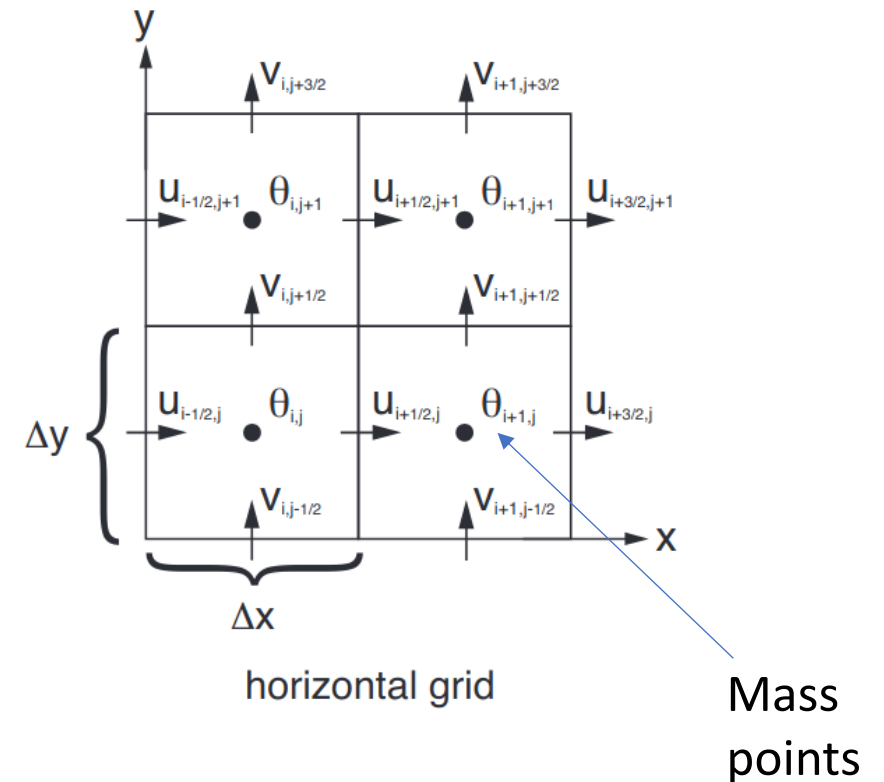


Fig. 3.2 of WRFv4 Technical notes

Introduction

- Flow over a hill
- WRF Model: Spatial discretization

The spatial discretization in the ARW solver uses a C grid staggering for the variables.

The vertical grid length $\Delta\eta$ is not a fixed constant, it is specified in the initialization. The user is free to specify the η values of the model levels subject to the constraint that $\eta = 1$ at the surface, $\eta = 0$ at the model top, and η decreases monotonically between the surface and model top.

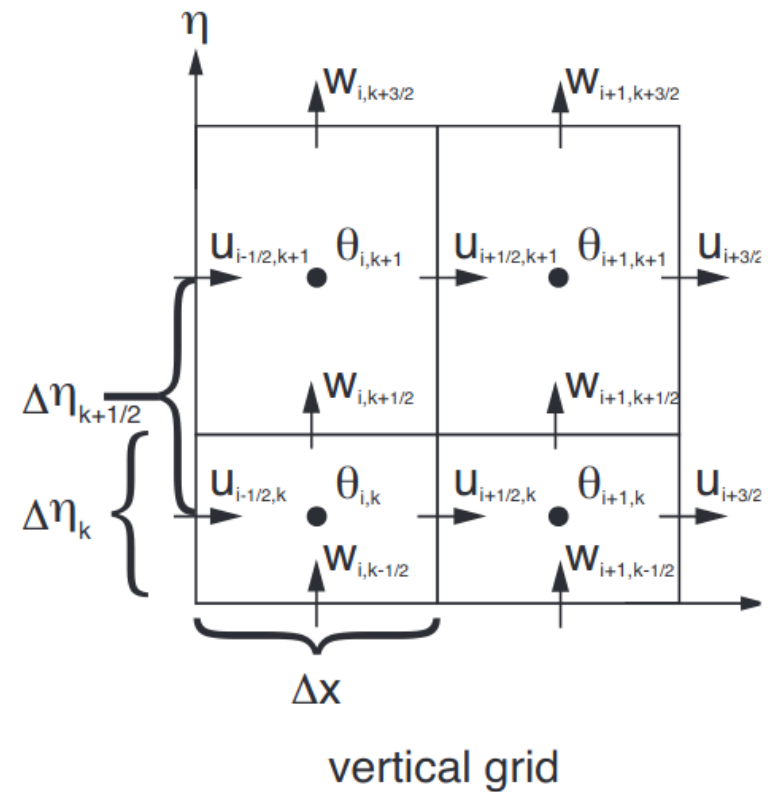
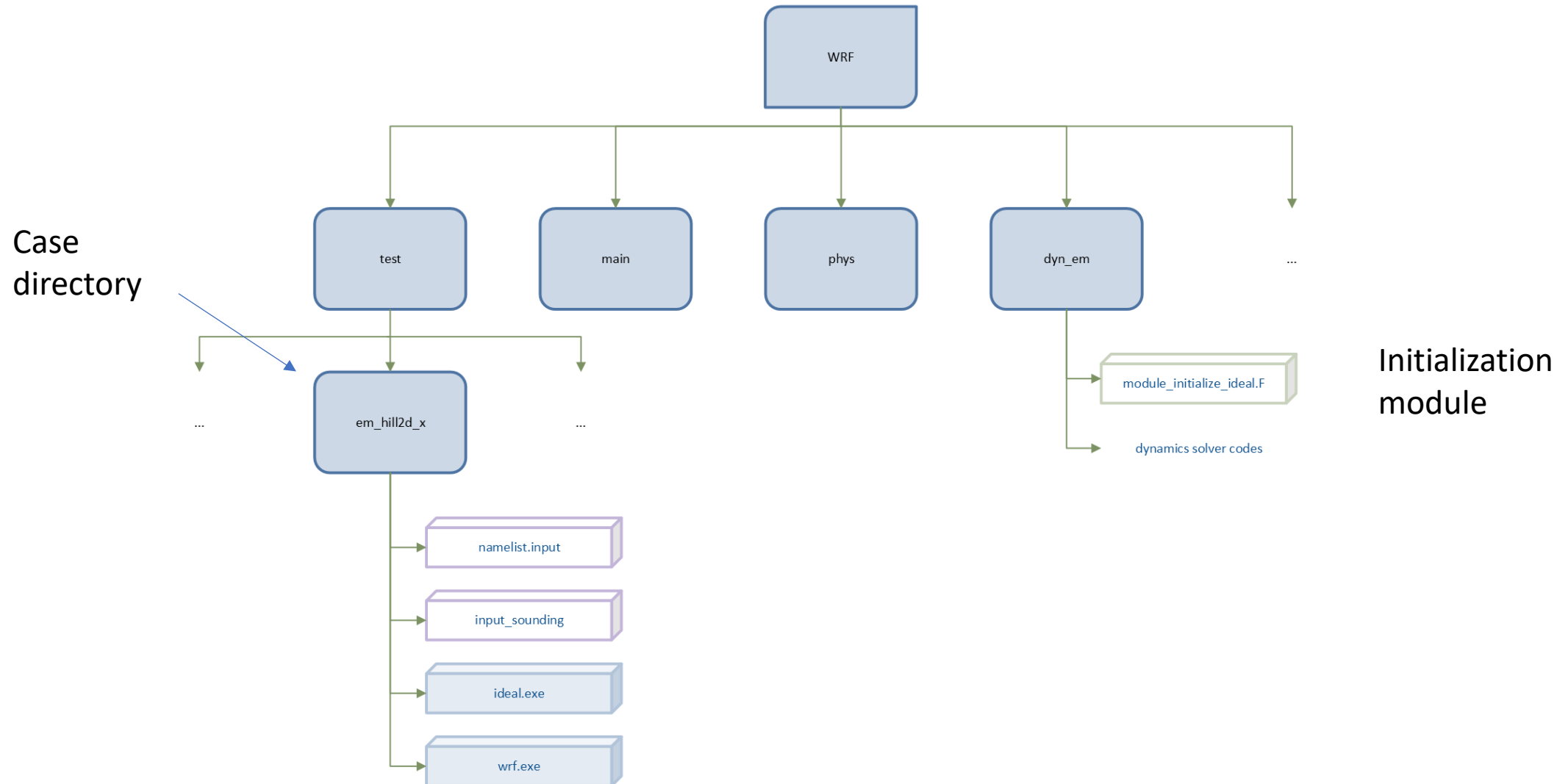


Fig. 3.2 of WRFv4 Technical notes

Introduction

- Flow over a hill • WRF Model: 2D hill simulation



Introduction

- Flow over a hill • WRF Model

Let's now see the case we will work on.

Steps:

- 1) Access the cluster
- 2) Set the environment: create a conda environment and install Ncview to view wrf outputs
- 3) Configure and compile WRF
- 4) Run the case

Introduction

- Flow over a hill • WRF Model

Access the cluster:

```
ssh -X -J nome.cognome@137.204.165.34 nome.cognome@137.204.165.41
```

Download WRF:

```
git clone --recurse-submodules https://github.com/wrf-model/WRF
```

To compile WRF use an interactive job:

```
salloc -N 1 --cpus-per-task=4 --time=00:30:00 --mem=32G --  
constraint=blade --partition=b1  
srun --pty --overlap --jobid $JOBID bash
```

Then, set the environment:

Introduction

- Flow over a hill
- WRF Model: Configuration and Compiling

```
tcsh
setenv DIR /home/software/atmos/WRFv4/Build_WRF/LIBRARIES
setenv CC gcc
setenv CXX g++
setenv FC gfortran
setenv FCFLAGS -m64
setenv F77 gfortran
setenv FFLAGS -m64
setenv JASPERLIB $DIR/grib2/lib
setenv JASPERINC $DIR/grib2/include
setenv LDFLAGS -L$DIR/grib2/lib
setenv CPPFLAGS -I$DIR/grib2/include
setenv PATH $DIR/netcdf/bin:$PATH
setenv NETCDF $DIR/netcdf
setenv LD_LIBRARY_PATH /home/software/atmos/WRFv4/Build_WRF/LIBRARIES/grib2/lib
setenv LD_LIBRARY_PATH $DIR/grib2/include:$LD_LIBRARY_PATH
module load mpi/openmpi/4.1.4
```

Introduction

- Flow over a hill
- WRF Model: Configuration and Compiling

In the WRF main directory type:

```
./configure
```

For idealized simulations, choose option 32 with no nesting (0)

Open `configure.wrf` file and modify line 137:

```
vi configure.wrf
```

type `:137` and press enter. Press `i` to go in `INSERT MODE` and delete “`time`”.

To exit press `esc` and save (`:wq`).

Now you can compile:

```
./compile em_hill2d_x >& compile.log
```

Call at the end:

```
scancel $JOBID
```

If compilation is successful, you should see executables in the main directory. In WRF folder, type:

```
ls main/
```

These commands refers to Vim text editor, but you can choose whatever you prefer (e.g. GNU nano)!

Introduction

- Flow over a hill • WRF Model: 2D hill simulation

dyn_em/module_initialize_ideal.F

```
CALL model_to_grid_config_rec ( grid%id , model_config_rec , config_flags )

ideal_constants: SELECT CASE ( model_config_rec%ideal_case )
CASE ( hill2d_x )
  hm = 100.
  xa = 5.0
  icm = ide/2

  xa1 = 5000./500.
  xa11 = 4000./500.
  pii = 2.*asin(1.0)
  hm1 = 250.
  hm1 = 1000.

  stretch_grid = .true.
  deltat = 0.
  z_scale = .40
  z_scale = 8000./config_flags%ztop
  pi = 2.*asin(1.0)
  write(6,*) ' pi is ',pi
  nxc = (ide-ids)/2
  nyc = (jde-jds)/2
```

hill height (m) and half-width (grid points)

hill position in domain (center gridpoint in x)

In vim:

Type `/hill`

Press `n` (`N`) to move forward (back) in the search

Introduction

- Flow over a hill • WRF Model: 2D hill simulation

dyn_em/module_initialize_ideal.F

```
CALL model_to_grid_config_rec ( grid%id , model_config_rec , config_flags )

ideal_constants: SELECT CASE ( model_config_rec%ideal_case )
CASE ( hill2d_x )
  hm = 100.
  xa = 5.0

  icm = ide/2

  xa1 = 5000./500.
  xa11 = 4000./500.
  pii = 2.*asin(1.0)
  hm1 = 250.
  hm1 = 1000.

  stretch_grid = .true.
  delt = 0.
  z_scale = .40
  z_scale = 8000./config_flags%ztop
  pi = 2.*asin(1.0)
  write(6,*) ' pi is ',pi
  nxc = (ide-ids)/2
  nyc = (jde-jds)/2
```

hill shape

lower bc

In vim:

Type `/hill`

Press `n` (`N`) to move
forward (back) in
the search

Witch of Agnesi

$$h(x) = \frac{h_m}{(x/b)^2 + 1}$$

Gaussian
shape

```
ideal_terrain: SELECT CASE ( model_config_rec%ideal_case )
CASE ( hill2d_x )
DO j=jts,jte
DO i=its,ite
  grid%ht(i,j) = hm/(1.+(float(i-icm)/xa)**2)
!!  grid%ht(i,j) = hm1*exp(-(( float(i-icm)/xa1)**2)) &
!!                    *( (cos(pii*float(i-icm)/xa11))**2 )
  grid%phb(i,1,j) = g*grid%ht(i,j)
  grid%php(i,1,j) = 0.
  grid%ph0(i,1,j) = grid%phb(i,1,j)
ENDDO
ENDDO
```

Introduction

- Flow over a hill • WRF Model: 2D hill simulation

namelist.input

Section (&)

Simulation time

Frequency (in min)
that data are written
in the wrfout* file

number of history
output times in
each wrfout* file

```
&time_control  
run_days = 0,  
run_hours = 10,  
run_minutes = 0,  
run_seconds = 0,  
start_year = 0001,  
start_month = 01,  
start_day = 01,  
start_hour = 00,  
start_minute = 00,  
start_second = 00,  
end_year = 0001,  
end_month = 01,  
end_day = 01,  
end_hour = 10,  
end_minute = 00,  
end_second = 00,  
history_interval = 60,  
frames_per_outfile = 1000,  
restart = .false.,  
restart_interval = 1440,  
io_form_history = 2  
io_form_restart = 2  
io_form_input = 2  
io_form_boundary = 2  
/
```

More on namelist variables in
[WRF UG section on Namelist variables](#)

Introduction

- Flow over a hill • WRF Model: 2D hill simulation

namelist.input

Time step (s)

s: start

e: end

```
&bdy_control  
periodic_x      = .false.,  
symmetric_xs    = .false.,  
symmetric_xe    = .false.,  
open_xs         = .true.,  
open_xe         = .true.,  
periodic_y      = .true.,  
symmetric_ys    = .false.,  
symmetric_ye    = .false.,  
open_ys         = .false.,  
open_ye         = .false.,  
/  

```

```
&domains  
time_step      = 20,  
time_step_fract_num = 0,  
time_step_fract_den = 1,  
max_dom        = 1,  
s_we           = 1,  
e_we           = 202,  
s_sn           = 1,  
e_sn           = 3,  
s_vert         = 1,  
e_vert         = 41,  
dx             = 2000,  
dy             = 2000,  
ztop           = 30000.,  
/  

```

Number of
grid points

Number of
vertical levels

Resolution
in metres

Open boundaries (x)

Periodic bc in y

Height of
the top in
metres

Introduction

- Flow over a hill • WRF Model: 2D hill simulation

surface potential temperature (K)

surface vapor mixing ratio (g/kg)

surface pressure (mb)

Each successive
line is a point in
the sounding

height (m)

potential
temperature (K)

input_sounding

V (south-north)
velocity (m/s)

U (west-east)
velocity (m/s)

1000.000	288.00	0.00		
0.00	288.00	0.00	10.00	0.00
50.00	288.15	0.00	10.00	0.00
100.00	288.29	0.00	10.00	0.00
150.00	288.44	0.00	10.00	0.00
200.00	288.59	0.00	10.00	0.00
250.00	288.74	0.00	10.00	0.00
300.00	288.88	0.00	10.00	0.00
350.00	289.03	0.00	10.00	0.00
400.00	289.18	0.00	10.00	0.00
450.00	289.32	0.00	10.00	0.00
500.00	289.47	0.00	10.00	0.00
550.00	289.62	0.00	10.00	0.00
600.00	289.77	0.00	10.00	0.00
650.00	289.92	0.00	10.00	0.00
700.00	290.06	0.00	10.00	0.00
750.00	290.21	0.00	10.00	0.00
800.00	290.36	0.00	10.00	0.00
850.00	290.51	0.00	10.00	0.00
900.00	290.66	0.00	10.00	0.00
950.00	290.80	0.00	10.00	0.00
1000.00	290.95	0.00	10.00	0.00

Introduction

- Flow over a hill • WRF Model: Initialization of ideal case

During initialization:

- Idealize test specifics are enabled in the code using the Fortran CASE construct.

```
SELECT CASE ( model_config_rec%ideal_case )  
CASE ( hill2d_x )
```
- Model levels are set within the initialization to produce a stretched η coordinate (close to equally spaced z), or equally spaced η coordinate.
- Terrain is set in the initialization code
- A single sounding (z , q , Q_v , u and v) is read in from

```
WRF/test/em_hill2d_x/input_sounding
```
- Sounding is interpolated to the ARW grid, equation of state and hydrostatic balance used to compute the full thermodynamics state.
- Wind fields are interpolated to model η levels.

Introduction

- Flow over a hill • WRF Model

Note:

3D meshes are always used, even in 2D (x,z; y,z) cases. The third dimension contains only 5 planes, the boundary conditions in that dimension are periodic, and the solutions on the planes are identical in the initial state and remain so during the integration.

Introduction

- Flow over a hill • WRF Model: 2D hill simulation

After compiling, we are ready to run the case. First, copy the bash scripts to your folder:

```
cp /scratch/atmos/simona.rinaldi7/complex_terrain_lab/run_* .
```

Set initial condition by running `ideal.exe`:

```
sbatch run_ideal.sh
```

Check `ideal.log` file:

```
wrf: SUCCESS COMPLETE IDEAL INIT
```

Then run wrf via `run_wrf.sh`

```
sbatch run_wrf.sh
```

After simulation has finished you should see a `wrfout*` file. Check `wrf.log` file

```
wrf: SUCCESS COMPLETE WRF
```

To open the output file with `ncview` activate conda environment:

```
conda activate env_name  
ncview wrfout*
```

To visualize all the variables available in a netcdf file

```
ncdump -h wrfout
```

Here it is required to have installed miniconda, created a conda environment and installed necessary packages (**nccl**, **ncview**). If this is not the case, go to the end of this presentation to see how to do that.

Introduction

- Flow over a hill • WRF Model

Froude number
Measures the degree of linearity

$$Fr_h = \frac{U}{Nh}$$

$Fr_h \gg 1$: linear flow

Aspect ratio
Measures the steepness of
flow streamlines

$$\beta = \frac{h}{b}$$

Measure of the hydrostaticity:

$$\frac{Nb}{U}$$

$\frac{Nb}{U} \gg 1$: hydrostatic limit

Nondimensional mountain height

$$\hat{h} = \frac{Nh}{U}$$

$\hat{h} \ll 1$: linear flow

U is the x-wind component
 N is the Brunt-Vaisala frequency
 h is the height of the mountain
 b is the half-width of the mountain

Part 0 – test case em_hill2d_x

Objective:

- Become familiar with the base-case input files and simulation output;
- Run the base case and interpret the results (calculate the parameters shown on Slide 29).

Part 0 – test case em_hill2d_x

Domain setup

$$L_x = 402 \text{ km}$$

$$\Delta x = \Delta y = 2 \text{ km}$$

41 vertical levels

$$U = 10 \text{ m/s}$$

$$N = 0.01 \text{ 1/s}$$

$$h = 100 \text{ m}$$

$$2b = 20 \text{ km}$$

- Linear parameter

$$\hat{h} = Nh/U = 0.1$$

Linear solution

- Hydrostatic parameter

$$\hat{b} = \frac{Nb}{U} = 10 \gg 1$$

Hydrostatic mountain waves

Part 0 – test case em_hill2d_x

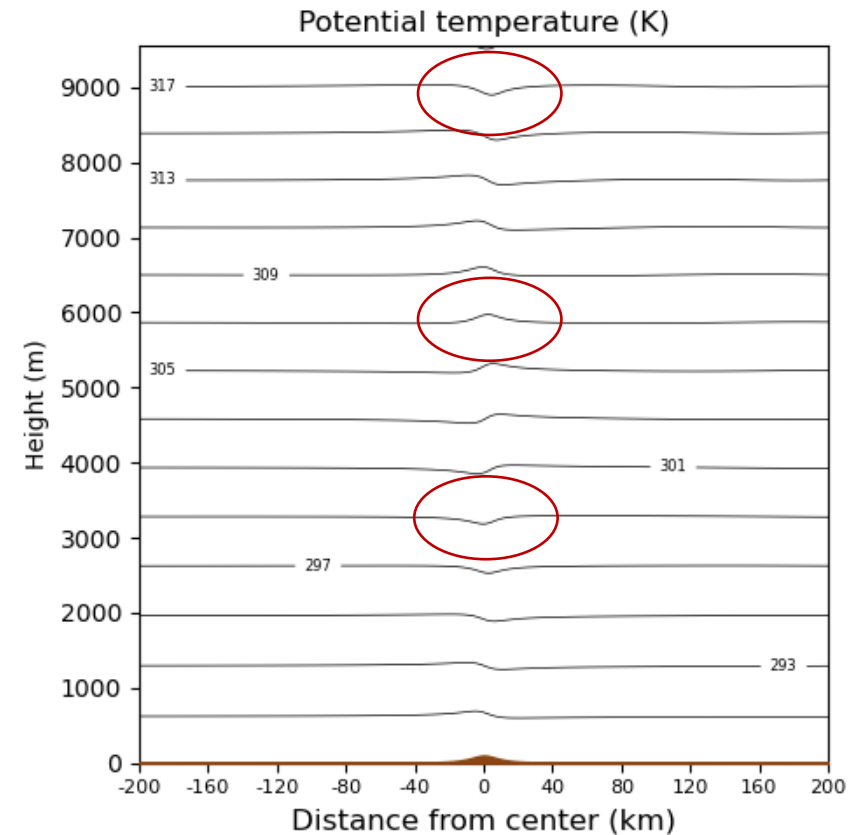
$$Nh/U = 0.1$$

Hydrostatic mountain waves

Isentropes are displaced upward and downward in a repeating vertical pattern with a wavelength of about 6 km.

The mountain profile is reproduced at every level that is an integral multiple of the wavelength λ , while those at 3 and 9 km are roughly the mirror-image of the topography.

$$\lambda = \frac{2\pi U}{N} \cong 6km$$

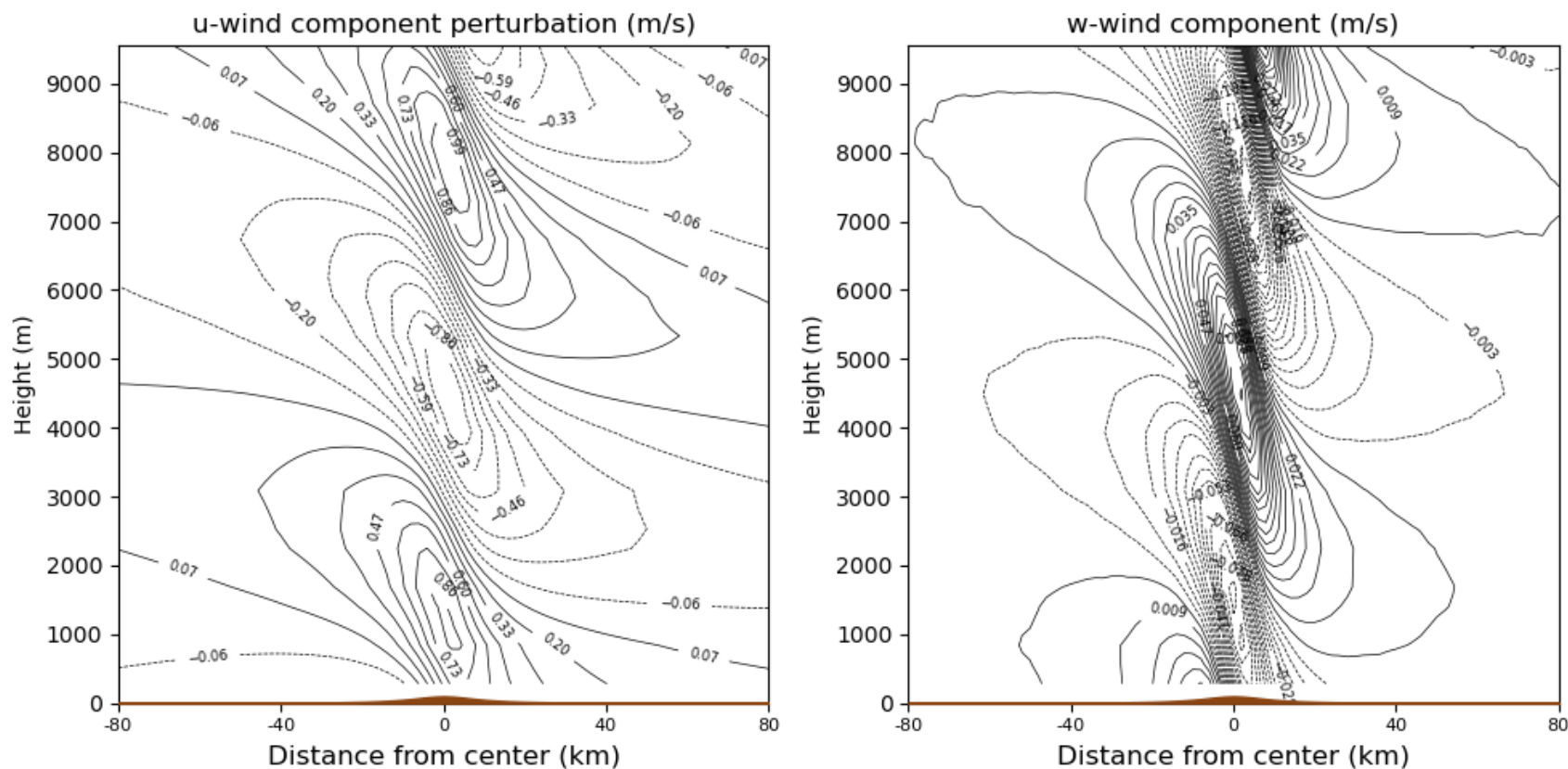


Cambio di segno con quota, vicino alla collina il flusso sta facendo downslope a valle che non è vero downslope. Acc molto piccola e non come venti di pendio.
 Altre condizioni per generare regimi ondosi maggiori. Se non idrostatico campo inizia a espandersi orizz e si vedono oscillazioni downstream, non solo sopra ridge.
 Campo d'onda si sviluppa downstream. In altre cond, trapped lee waves: Lee è lato sotto-vento. Cond necessaria e non suff è che fluido abbia variaz verticale, quindi shear e non cost
 o N non cost con quota. Se discount, la linea di discount è un riflettore e onde non passano ma si amplificano nello strato con interferenza costruttiva.

Part 0 – test case em_hill2d_x

$$Nh/U = 0.1$$

Hydrostatic mountain waves



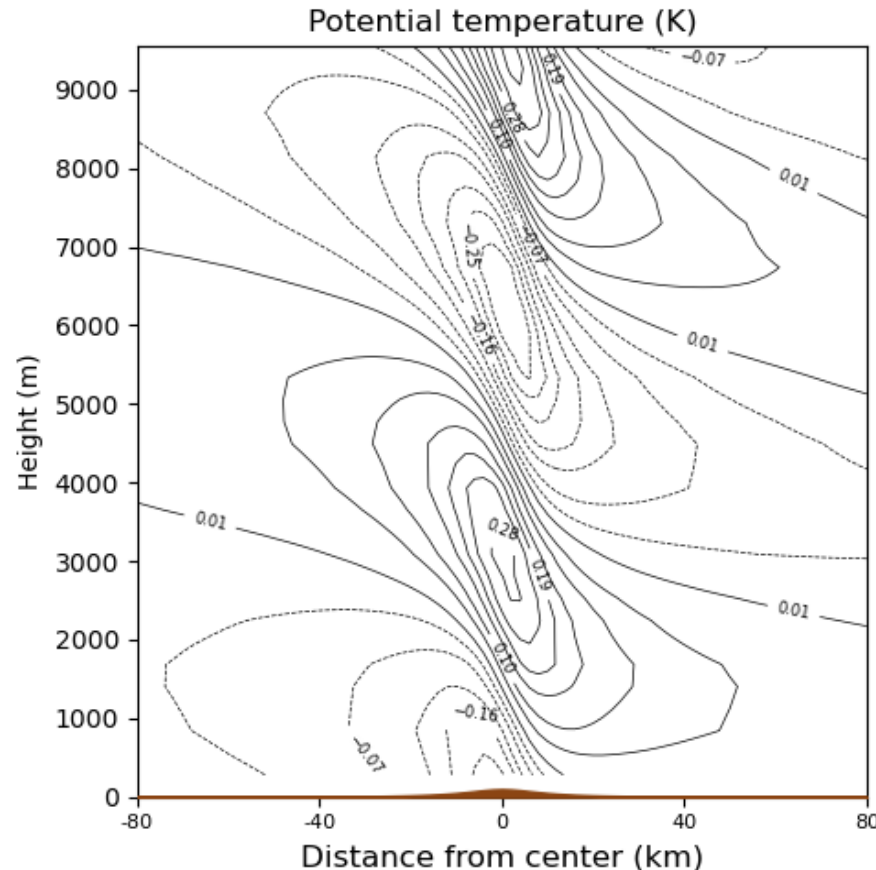
Part 0 – test case em_hill2d_x

$$Nh/U = 0.1$$

Hydrostatic mountain waves

The hill forces rising and sinking motions in a stably stratified flow, generating stationary gravity waves.

The disturbance is confined in the horizontal above the mountain, but repeats itself in vertical.



Part 1.1 – 2-layer atmosphere

Objective: investigation of the development of **trapped lee waves**.

In this case, the atmosphere has two layers, with the upper layer being less stable than the bottom one.

Case	N_L (s^{-1})	N_U (s^{-1})	H (m)
1	0.02	0.01	1571
2	0.02	0.01	3142
3	0.01	0.02	3142
4	0.01	0.02	6243

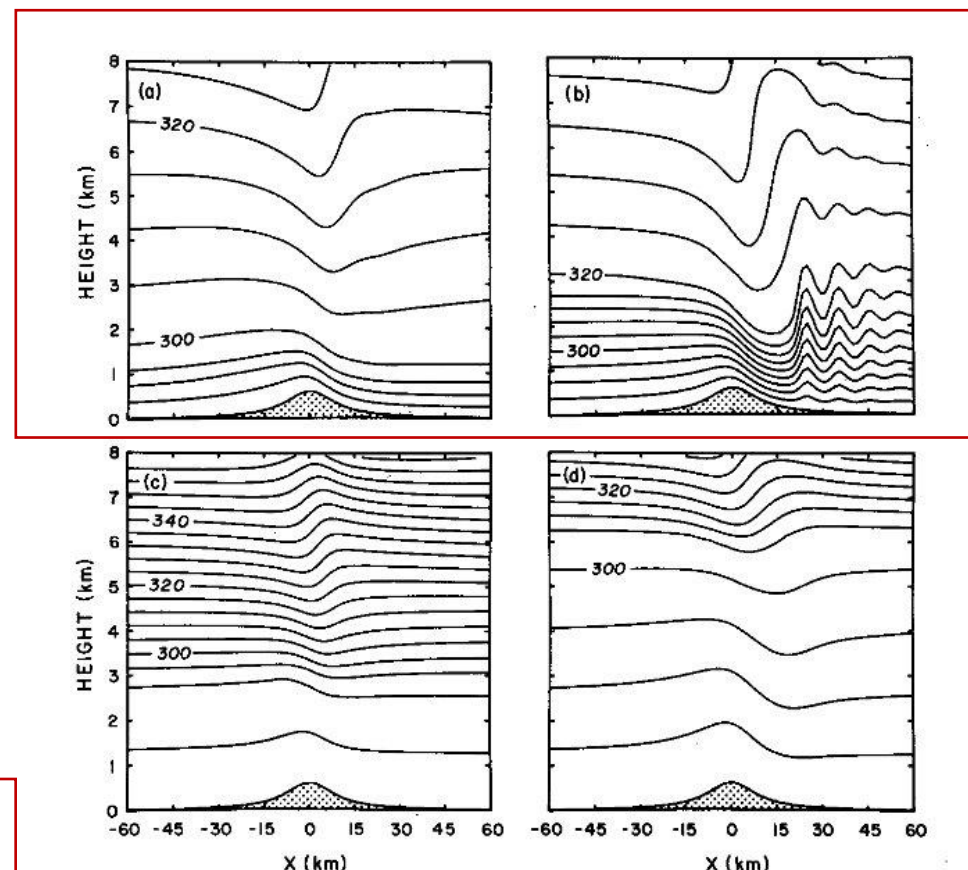


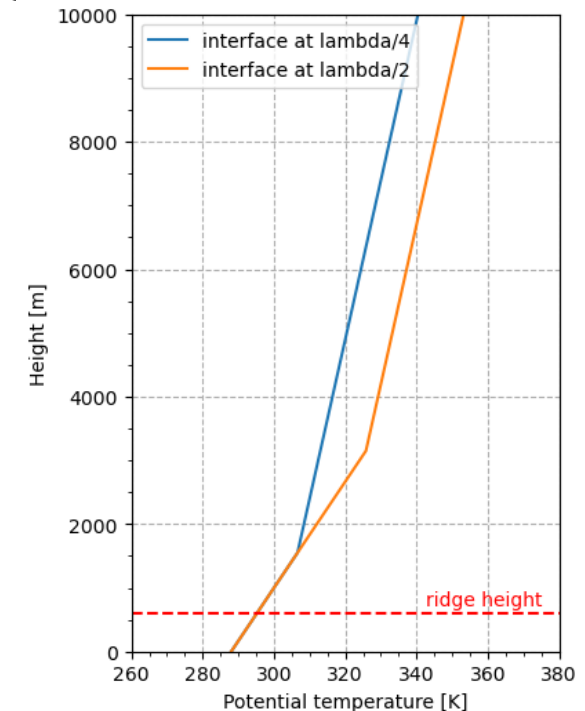
FIG. 1. Isentropes for the air in a two-layer atmosphere flowing over a 600 m high mountain at a nondimensional time $Ut/a = 20$. (a) case 1: interface at 1571 m, one-quarter wavelength; (b) case 2: interface at 3142 m, one-half wavelength; (c) case 3: interface at 3142 m, one-quarter wavelength; (d) case 4: interface at 6243 m, one-half wavelength. In this and all the figures which follow, the airflow is from left to right.

Part 1.1 – 2-layer atmosphere

Build the case:

1. Create a new WRF folder:
`cp -r /scratch/atmos/simona.rinaldi7/complex_terrain_lab/WRFv4.6/WRF .`
2. Compile WRF, changing ridge height to 600 m (see slide 20)
3. Construct the sounding with the following settings:
 - Interface at $\lambda = h/4$ and $\lambda = h/2$
 - Bottom layer with $N_L = 0.02$ 1/s and $N_U = 0.01$ 1/s
 - Wind speed $U = 20$ m/s

$$\lambda = \frac{2\pi U}{N_L}$$

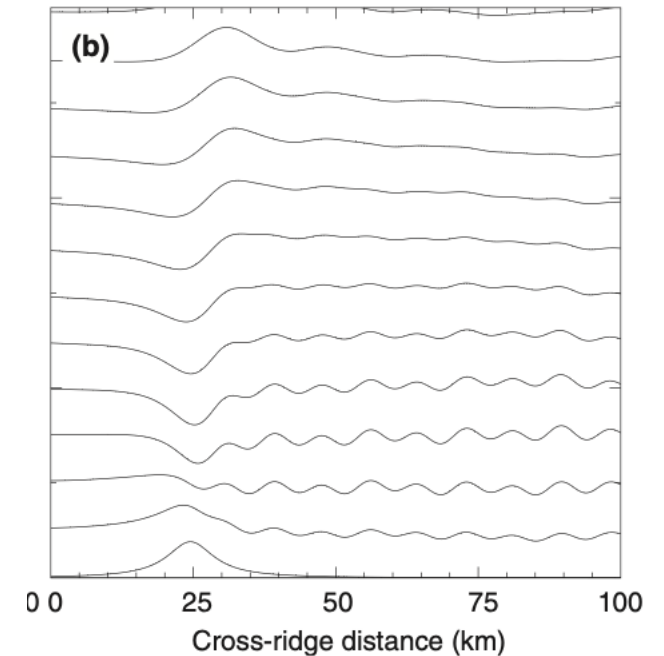


Part 1.1 – 2-layer atmosphere

If wind speed increases with height, or stability decreases with height, two layers with different Scorer parameter (l_U and l_L for the upper and lower layers) can form. Assuming $l_U < l_L$, waves satisfying $l_L > k > l_U$, will propagate vertically in the lower layer but decay with height in the upper layer. When the zonal wind or static stability are approximately constant within each layer, waves can experience refraction and/or reflection at the adjoining interface.

In this case, cross-ridge flow may generate waves that extend downstream from the ridge, known as **trapped lee waves**.

$$\ell_L^2 - \ell_U^2 > \frac{\pi^2}{4H^2},$$

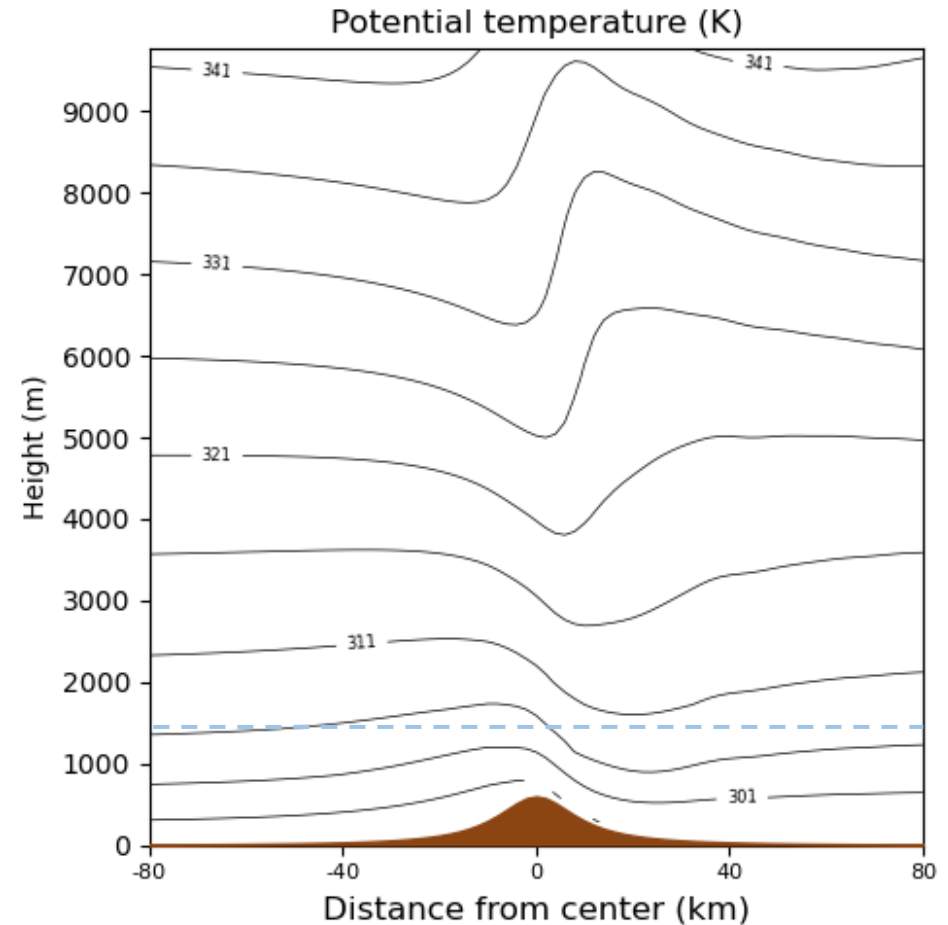


Streamlines in steady air flow over an isolated mountain as predicted by linear theory (Durran, 2015).

Part 1.1 – 2-layer atmosphere

Interface at $\lambda/4 \cong 1550m$

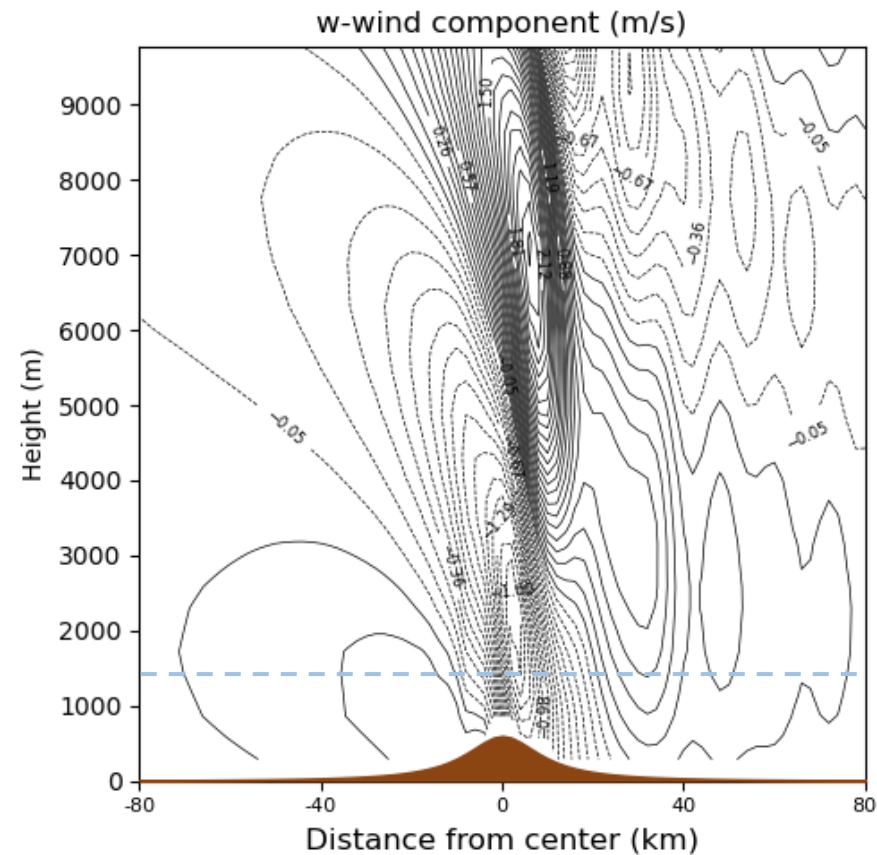
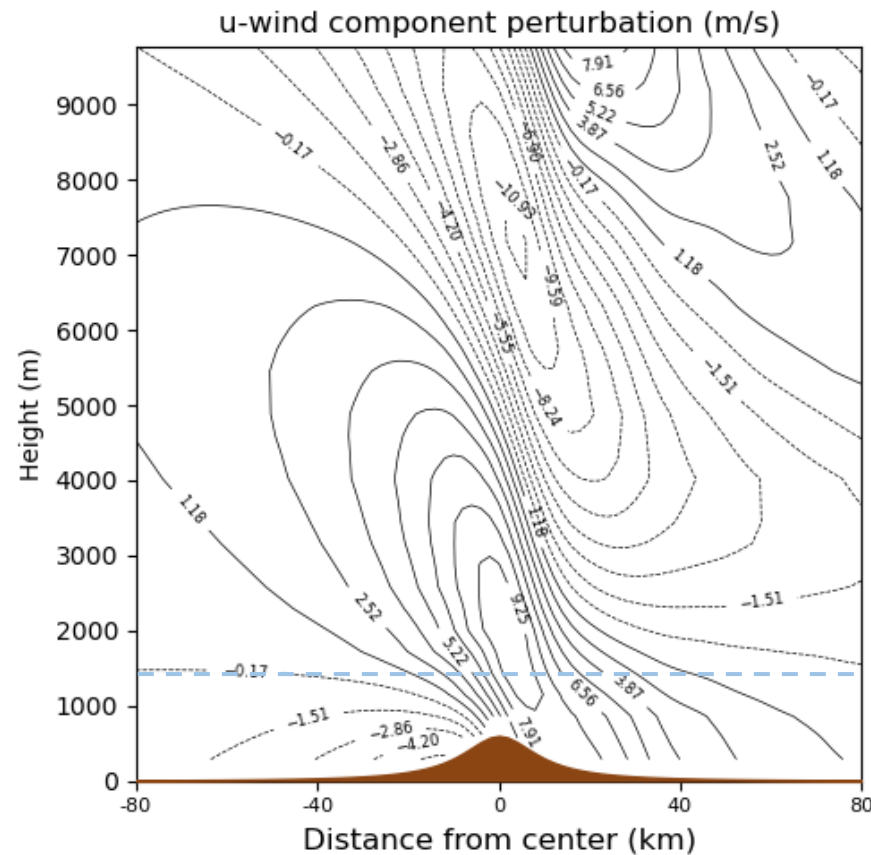
In this case, we have a single vertically propagating wave.



Solution at $\frac{Ut}{a} = 20$

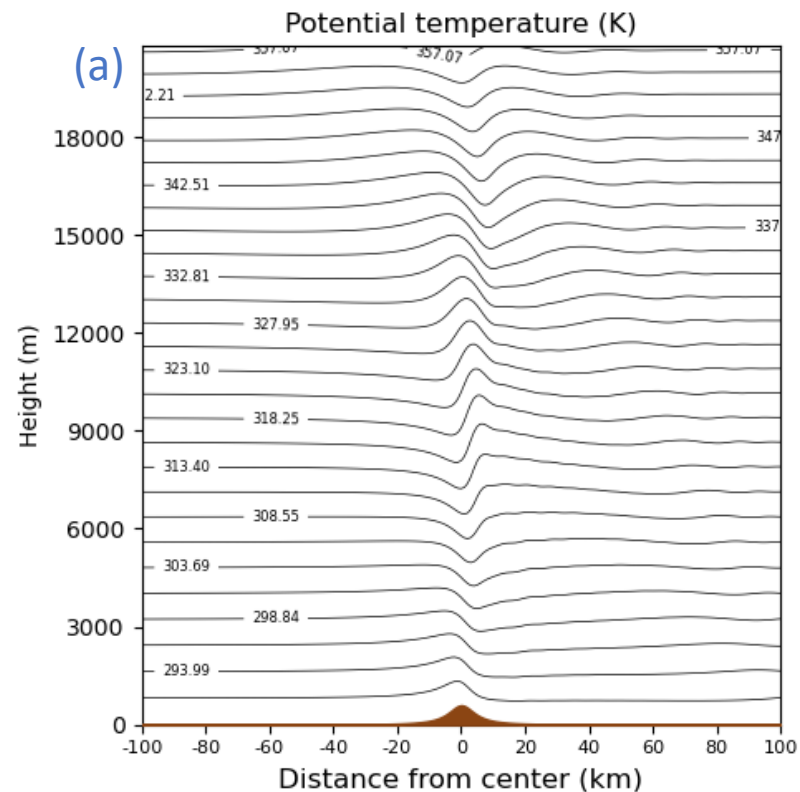
Part 1.1 – 2-layer atmosphere

Interface at $\lambda/4 \cong 1550m$

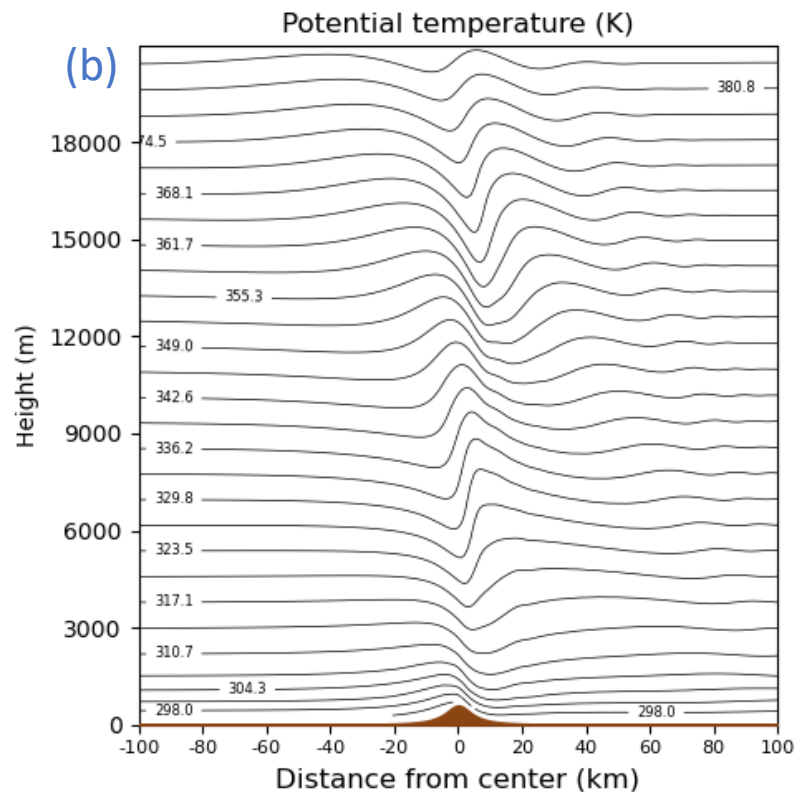


Part 1 – 2-layer atmosphere

Difference between one layer (a) and two-layer (b) atmosphere, where the one layer case has the same stability of the upper layer of case (b).



$$\hat{h} = Nh/U = 0.3$$



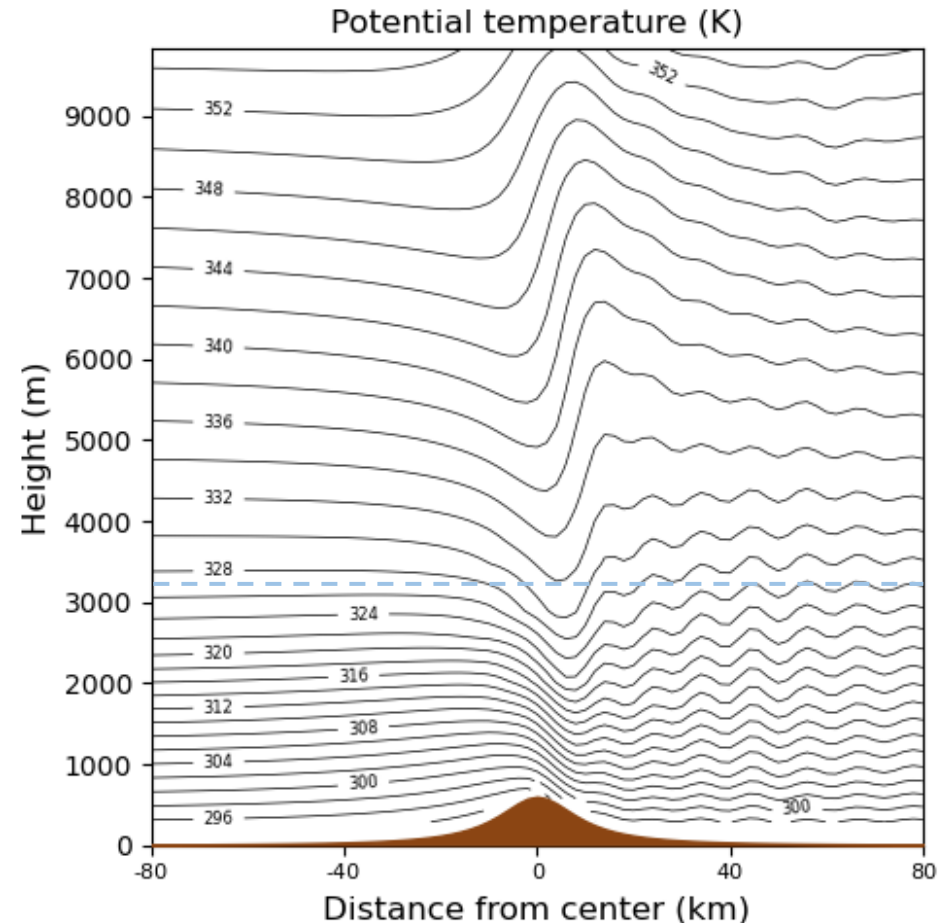
$$\hat{h}_U = Nh/U = 0.3$$

$$\hat{h}_L = Nh/U = 0.6$$

Part 1 – 2-layer atmosphere

Interface at $\lambda/2 \cong 3150m$

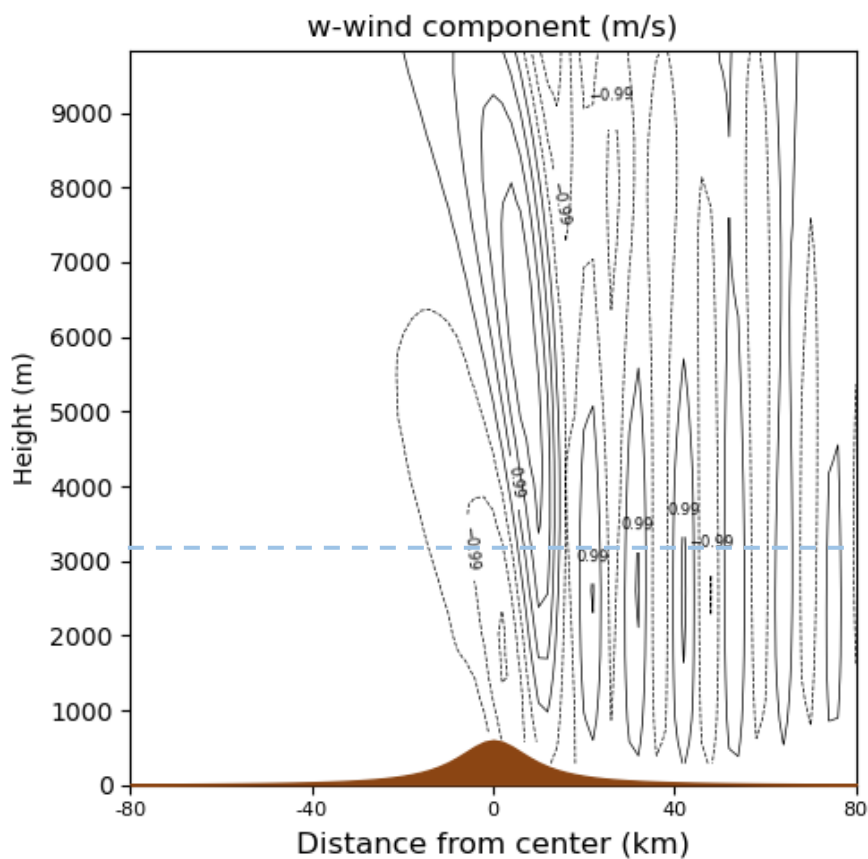
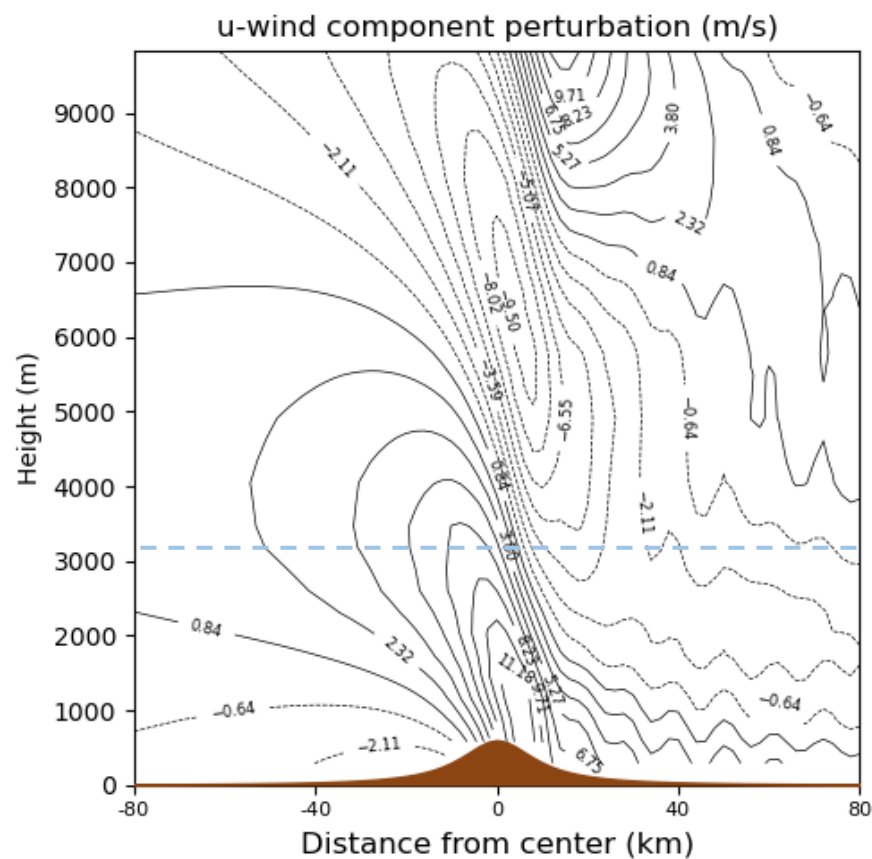
In this case, trapped lee waves occur. The horizontal wave number of any lee waves satisfies $l_L > k > l_U$, implying that the wave propagates vertically in the lower layer and decays exponentially with height in the upper layer.



Solution at $\frac{Ut}{a} = 20$

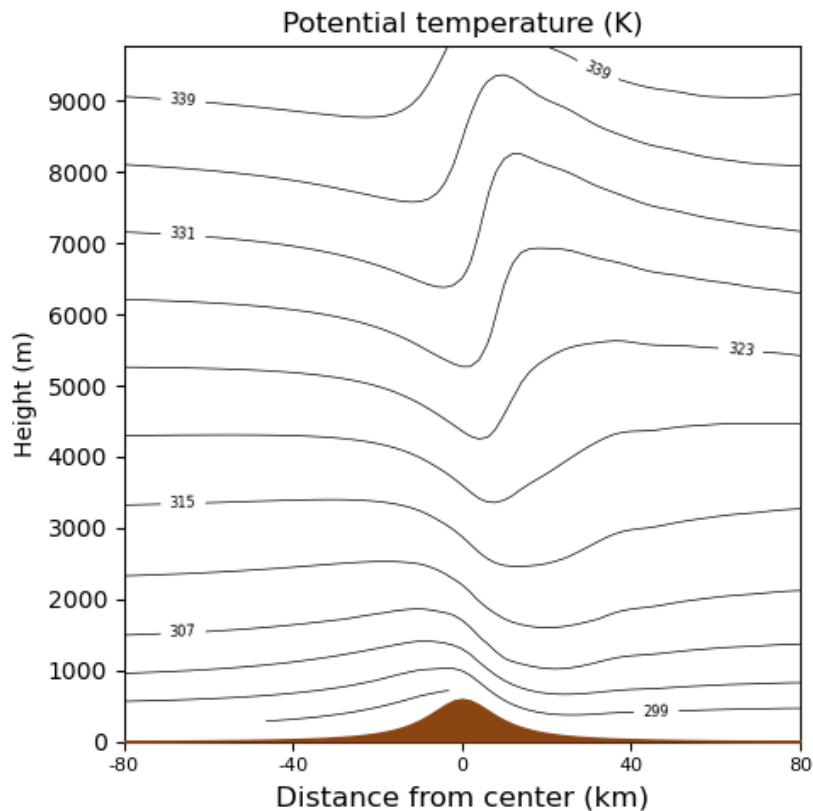
Part 1 – 2-layer atmosphere

Interface at $\lambda/2 \cong 3150m$

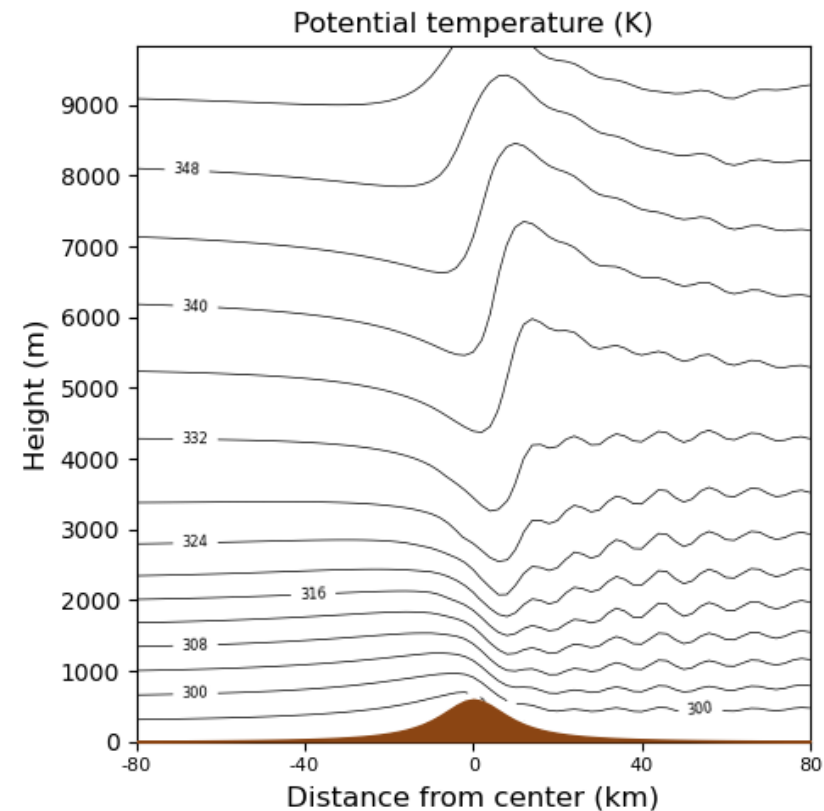


Part 1 – 2-layer atmosphere

Difference between the different position of the interface



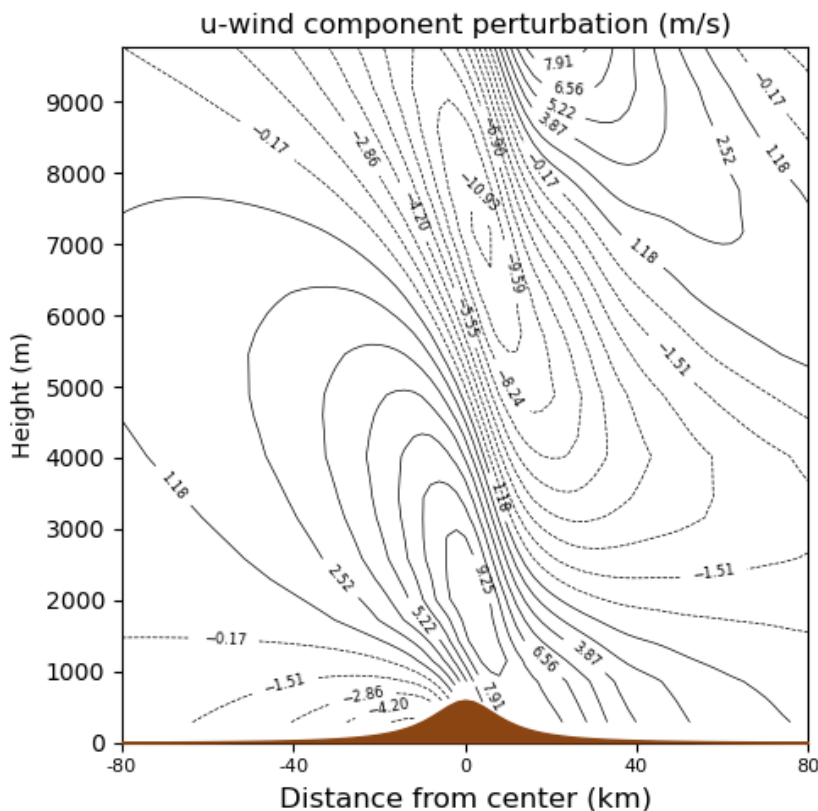
$$z_i = \lambda/4 \cong 1550\text{m}$$



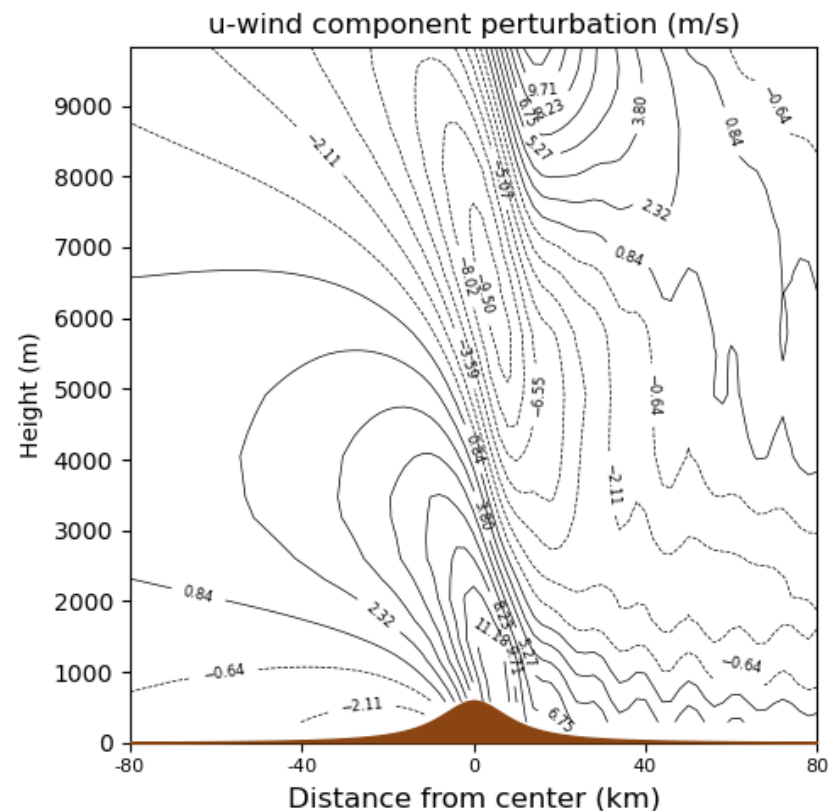
$$z_i = \lambda/2 \cong 3150\text{m}$$

Part 1 – 2-layer atmosphere

Difference between the different position of the interface



$$z_i = \lambda/4 \cong 1550m$$



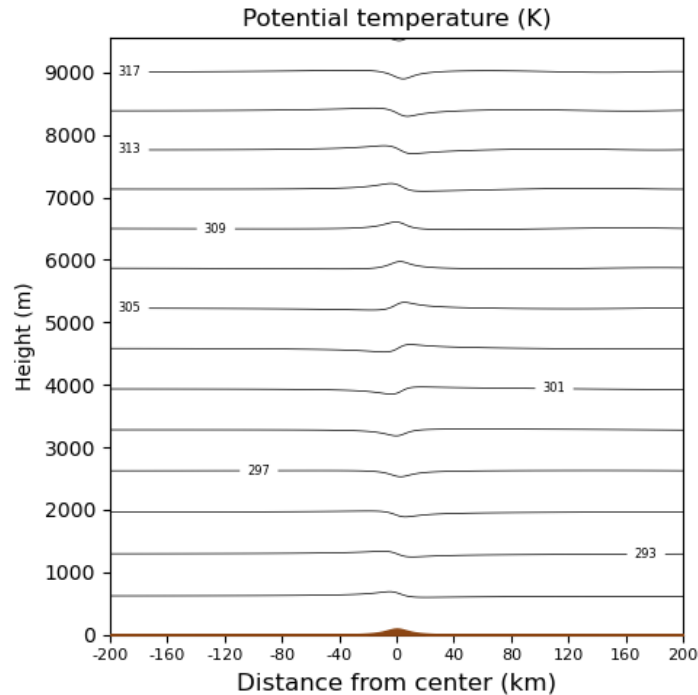
$$z_i = \lambda/2 \cong 3150m$$

Part 1.2 – non-linear effects

1. Copy the WRF folder pre-compiled with hill height at 1000m:
`cp -r /scratch/atmos/simona.rinaldi7/complex_terrain_lab/WRF_hill1000 .`
2. Set the input sounding to have a nonlinear case (e.g. $\hat{h} = 1$)

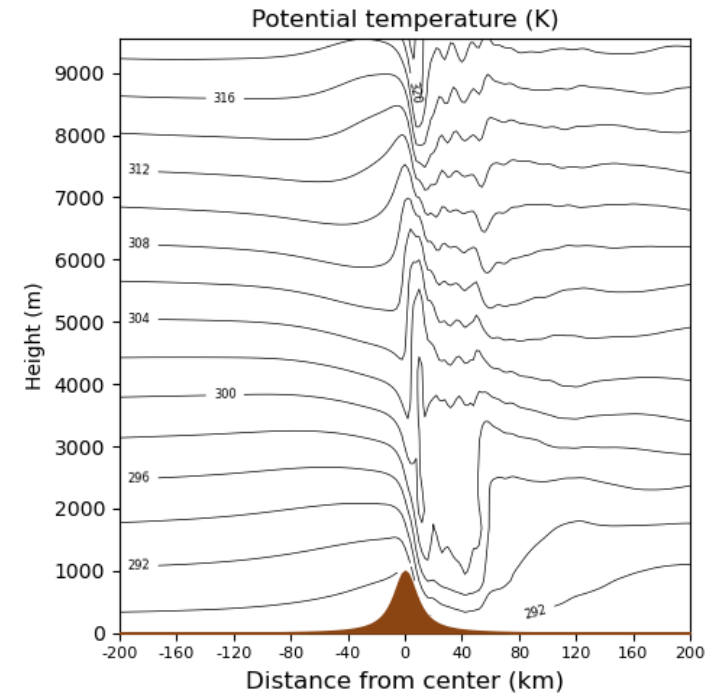
Part 1.2 – non-linear effects

$$\begin{aligned} h &= 100 \text{ m} \\ U &= 10 \text{ m/s} \\ N &= 0.01 \text{ 1/s} \end{aligned} \quad \begin{aligned} \hat{h} &= 0.1 \\ \hat{b} &= 10 \end{aligned}$$



Linear case

$$\begin{aligned} h &= 1000 \text{ m} \\ U &= 10 \text{ m/s} \\ N &= 0.01 \text{ 1/s} \end{aligned} \quad \begin{aligned} \hat{h} &= 1 \\ \hat{b} &= 10 \end{aligned}$$



Non-linear case

Part 1.2 – non-linear effects

Domain setup

$$L_x = 402 \text{ km}$$

$$2b = 20 \text{ km}$$

$$\Delta x = \Delta y = 2 \text{ km}$$

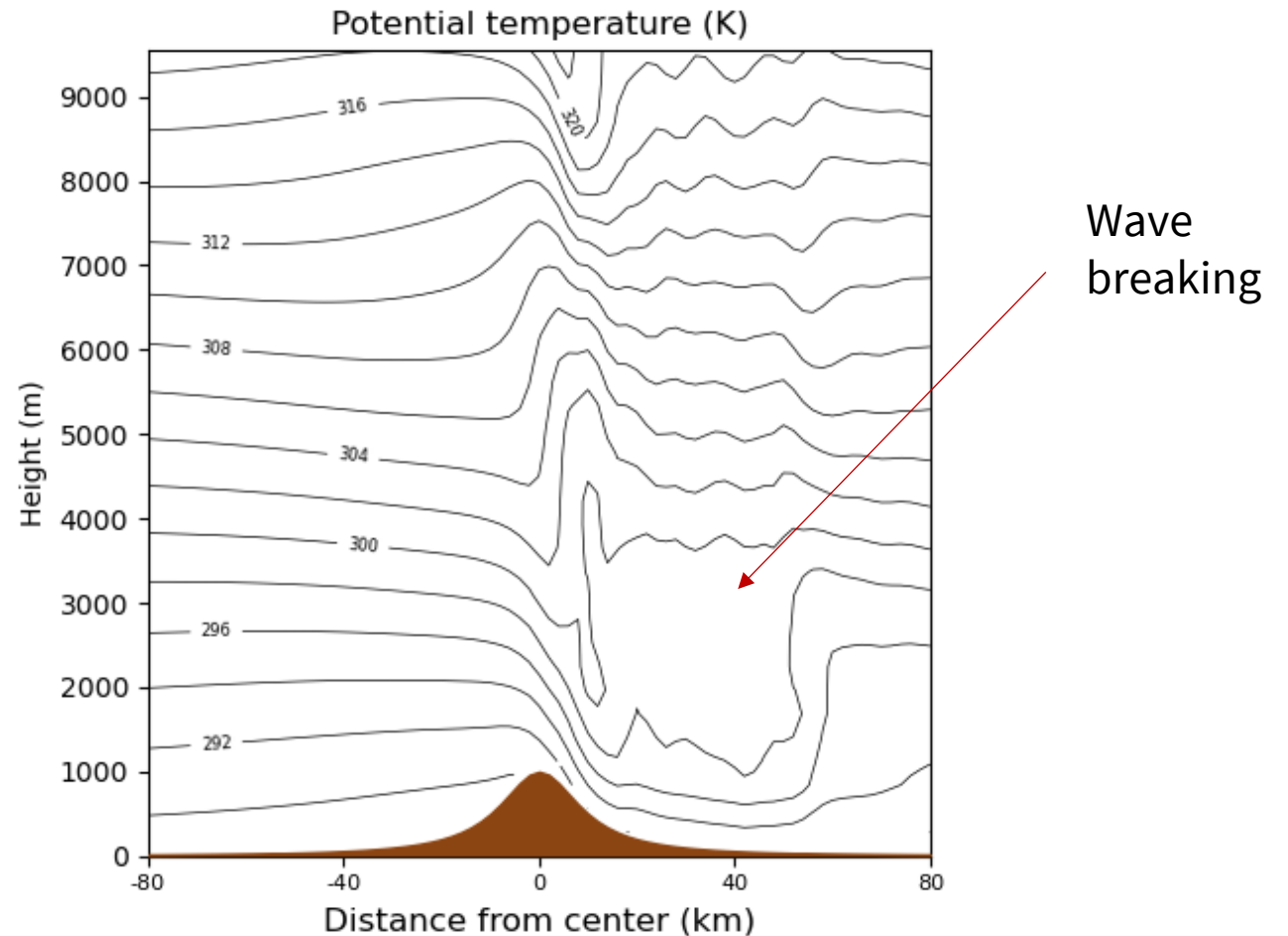
41 vertical levels

$$U = 10 \text{ m/s}$$

$$N = 0.01 \text{ 1/s}$$

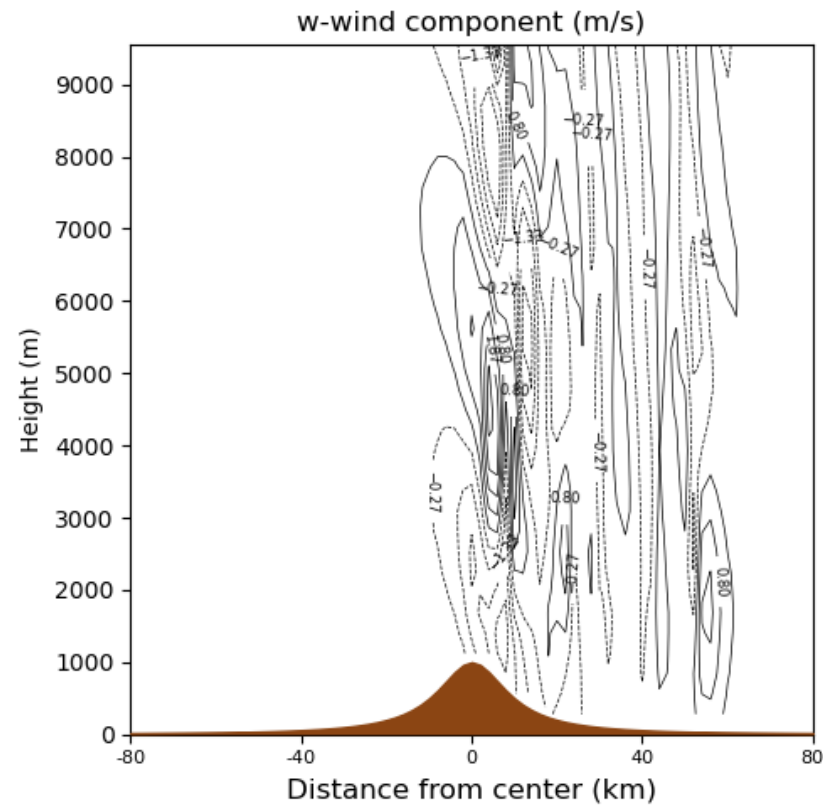
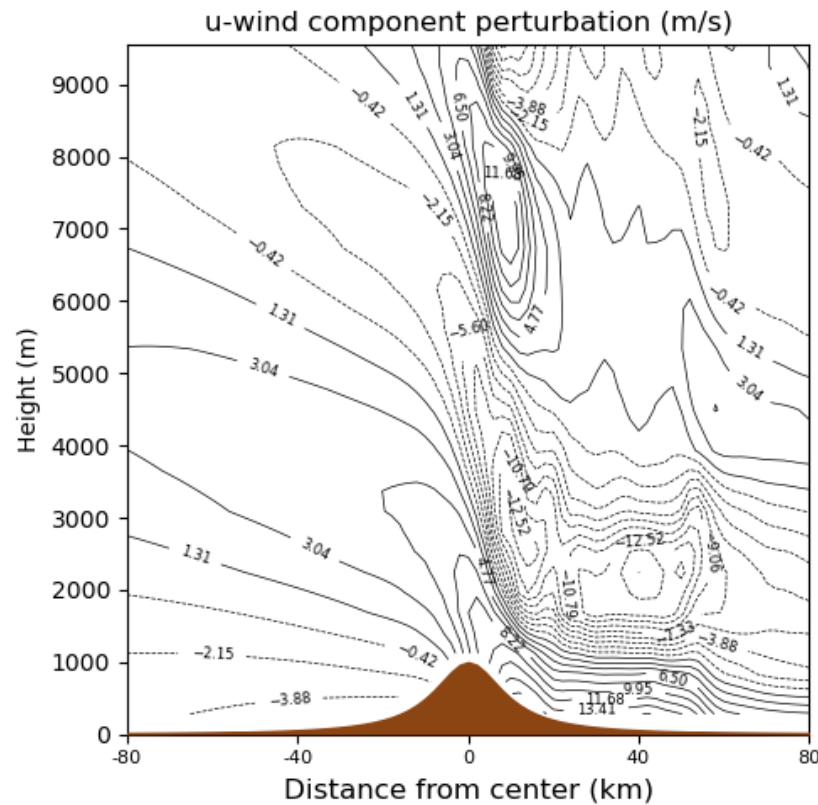
$$h = 1000 \text{ m}$$

$$\hat{h} = \frac{Nh}{U} = 1$$



Part 1.2 – non-linear effects

h=1000m, U=10 m/s



Soundings and Plotting

- How to change the sounding? One way is to use an [excel worksheet](#).
- For plotting we can use [wrf-python](#).

In VSCode, follow this [guide](#) to see how to create a new Jupyter notebook:

- > Create a new Jupyter Notebook: ctrl+shift+p (or creating a .ipynb file)
- > Select a kernel (top right)
- > Create a conda environment (with a python version ≤ 3.11)
- > Install libraries (In a code cell type: `%conda install metpy`)

Terminal commands

- Access the cluster

`ssh -Y -J nome.cognome#@137.204.165.34 nome.cognome#@137.204.165.41`

- Request an interactive job (frontend)

`salloc -N 1 --cpus-per-task=4 --time=00:30:00 --mem=32G --constraint=blade --partition=b1`

`srun --pty --overlap --jobid $JOBID bash`

- Submit job via sbatch

`sbatch script_name.sh`

In here remember to modify the bash script and insert your email

- Control job queue

`squeue -u nome.cognome#`

or

`slurmtop`

To exit from slurmtop press `ctrl+c`

Terminal commands

Soft link a file

```
ln -sf name of the file softlink name
```

To access the cluster open another terminal window and insert:

```
ssh -Y -J nome.cognome#@137.204.165.34 nome.cognome#@137.204.165.41
```

To **download** a file from the cluster, go into the local folder and insert:

```
scp -J nome.cognomen#@137.204.165.34 nome.cognome#@137.204.165.41:path/of/the/file/ .
```

To **upload** a file in the cluster, go into the folder containing the file and type:

```
scp -J nome.cognomen#@137.204.165.34 filename nome.cognome#@137.204.165.41:pathofthefinalfolder
```

Terminal commands

To mount remote filesystem, firstly install sshfs:

```
sudo apt-get install sshfs
```

Then, create a local mount point (where you want to have the cluster folder):

```
mkdir ~/folder_name
```

Now you can mount the remote directory:

```
sshfs -o ProxyJump=nome.cognome#@137.204.165.34 nome.cognome#@137.204.165.41:/path/of/clusterfolder  
~/folder_name
```

To unmount the folder, type in the directory of the mount point:

```
umount -f folder_name
```

If unmounting is unsuccessful, force it:

```
diskutil unmount force folder_name
```

Create a conda environment in the cluster

In the cluster

Firstly, install [miniconda](#). Go into main folder and type:

```
mkdir -p ~/miniconda3
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O
~/miniconda3/miniconda.sh
bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3
rm ~/miniconda3/miniconda.sh
```

Then, you can create a conda environment:

```
conda create -n env_name python=3.8
```

To activate the environment:

```
conda activate env_name
```

To install packages, activate conda environment first and then install:

```
conda install -c conda-forge package_name
```

Install [nccl](#) and [ncview](#).

Create a conda environment in the cluster

For macOS (locally):

```
brew install --cask miniconda
```

Firstly, install [miniconda](#). Go into main folder and type:

```
mkdir -p ~/miniconda3
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O
~/miniconda3/miniconda.sh
bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3
rm ~/miniconda3/miniconda.sh
```

Then, you can create a conda environment:

```
conda create -n env_name python=3.8
```

To activate the environment:

```
conda activate env_name
```

To install packages, activate conda environment first and then install:

```
conda install -c conda-forge package_name
```

Install [ncl](#) and [ncview](#).

References

- Lin, Y., and T. Wang, 1996: Flow Regimes and Transient Dynamics of Two-Dimensional Stratified Flow over an Isolated Mountain Ridge. *J. Atmos. Sci.*, 53, 139–158, [https://doi.org/10.1175/1520-0469\(1996\)053<0139:FRATDO>2.0.CO;2](https://doi.org/10.1175/1520-0469(1996)053<0139:FRATDO>2.0.CO;2)
- Durran, D.R., 2015. Lee Waves and Mountain Waves. In: Gerald R. North (editor-in-chief), John Pyle and Fuqing Zhang (editors). *Encyclopedia of Atmospheric Sciences*, 2nd edition, Vol 4, pp. 95–102.
- Durran, D. R., 1986: Another Look at Downslope Windstorms. Part I: The Development of Analogs to Supercritical Flow in an Infinitely Deep, Continuously Stratified Fluid. *J. Atmos. Sci.*, **43**, 2527–2543, [https://doi.org/10.1175/1520-0469\(1986\)043<2527:ALADWP>2.0.CO;2](https://doi.org/10.1175/1520-0469(1986)043<2527:ALADWP>2.0.CO;2).
- Reinecke, P. A., and D. R. Durran, 2008: Estimating Topographic Blocking Using a Froude Number When the Static Stability Is Nonuniform. *J. Atmos. Sci.*, **65**, 1035–1048, <https://doi.org/10.1175/2007JAS2100.1>.
- Stull, R., 2017: "Practical Meteorology: An Algebra-based Survey of Atmospheric Science" -version 1.02b. Univ. of British Columbia. 940 pages. isbn 978-0-88865-283-6.

References

WRF-related:

[WRF Technical notes \(v4\)](#)

[WRF User Guide](#)

[WRF ARW User's Page](#)

[WRF GitHub Repository](#)

[Ncview features](#)

[NCL installation](#)

Other useful stuff:

[Vim cheatsheet](#)

[OPH Guide](#)

[Lab folder](#)

Introduction

- Flow over a hill • WRF Model

If needed WRF folder can be copied from:

```
cp -r /scratch/atmos/simona.rinaldi7/complex_terrain_lab/WRFv4.6/WRF .
```